Freight Rate Prediction — Technical Assessment

Scope and Data

Labeled development set: 48,000 loads dated January 1–October 31, 2025. Final validation set: 12,000 loads. December chart inputs: 31 daily rows. Target variable: posted_rate. The validation dataset has no target labels.

Data Quality and Preprocessing

The development data has 300 missing weight values and 374 missing market_index values. CatBoost handles missing numerical values natively. Categorical features are standardized as strings. Date fields are parsed and converted into numerical calendar features. Missing fields in December, including coordinates, market_index and quote_signal, are left missing rather than estimated. No labeled target data from validation or December is used during training.

Features and Model

The selected model is CatBoost Regression, using gradient-boosted decision trees. The configuration includes 550 iterations, tree depth 8, learning rate 0.055, and L2 regularization of 5.

Engineered features include pickup location, delivery location, equipment type, origin-destination lane, distance, squared distance, weight, weight per mile, geographical coordinates where available, market index, quote signal, interaction features, and calendar features.

CatBoost was selected because it supports mixed categorical and numerical data and handles missing numerical values.

Chronological Holdout Validation

The labeled development data was split chronologically to evaluate future-period prediction performance.

Training period: January–September 2025 (43,147 records). Holdout validation period: October 2025 (4,853 records).

A chronological split was selected instead of a random split because the task involves predicting future freight rates. This approach reduces temporal leakage and better represents real-world forecasting.

After evaluating the model, the same configuration was retrained using all 48,000 labeled records before generating the final predictions.

Model Evaluation Results

Metric	October 2025 Holdout

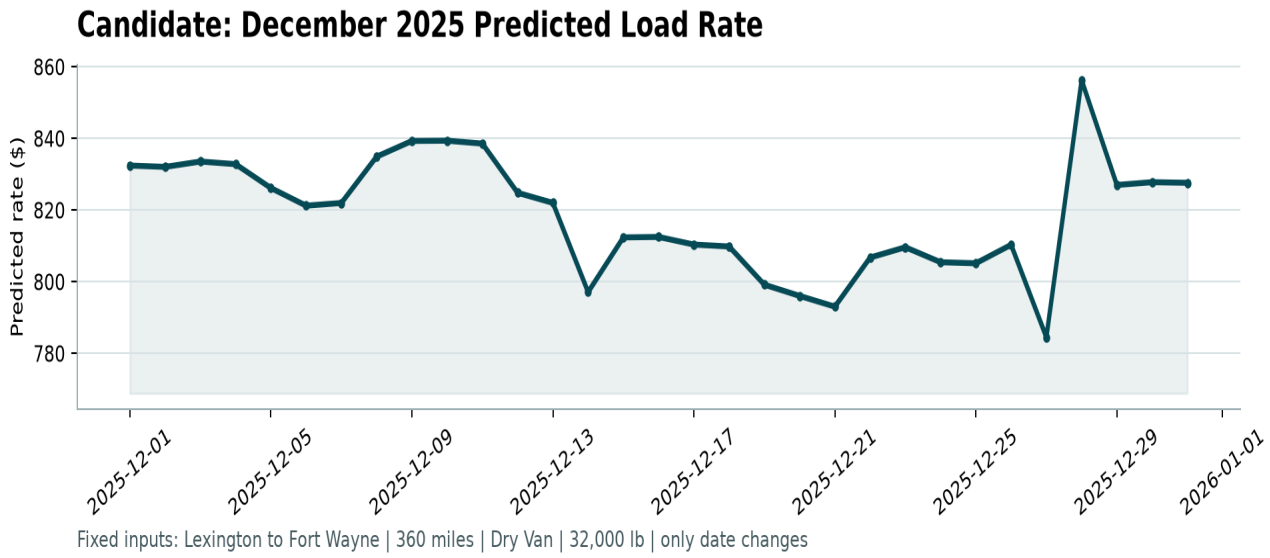
MAE	171.16
RMSE	655.09
R^2	0.8163
WAPE	7.19%

December Predictions — Official Scorer Chart

The following chart was generated by the official score.py script after validating all 12,000 final predictions and all 31 December predictions.

December inputs contain fewer features than the training and validation datasets. Therefore, prediction performance may differ from the October holdout results.

Final hidden validation metrics will be calculated by Spotter after submission.



Reproducibility

The project can be reproduced by installing the dependencies in requirements.txt, running train.py to train the model and generate predictions, and executing score.py to validate the output files.

Final validation predictions are matched to the required template using load_id and checked for completeness.